Model 3

```
library('rjags')
```

```
## Loading required package: coda
```

```
## Linked to JAGS 4.3.2
```

```
## Loaded modules: basemod,bugs
```

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(tibble)
library(stringr)
library(ggplot2)
library(tidyr)

output_folder = "../data/models/v3/"
save_output   = TRUE
if (save_output) dir.create(output_folder, recursive = TRUE, showWarnings = FALSE)
```

Preprocess

Load data Loading data in as a wide data frame

With ~ rows (regions) x columns (weeks)

1st column is the region names

```
uecDf    = read.csv('../data/wide_weekly_scaledPer10k.csv')
regions  = uecDf$Region # saving region names
n.region = length(uecDf$Region) # number of regions
uecDf    = uecDf %>% select(-Region) # remove region col
n.weeks  = length(names(uecDf))
head(uecDf)
```

```
##   X2023.01.01 X2023.01.08 X2023.01.15 X2023.01.22 X2023.01.29 X2023.02.05
## 1   0.9001159    6.857584    5.521329    5.289341    4.222193    4.491300
## 2   0.6823454    4.953323    3.285367    2.426117    2.021764    1.726924
## 3   0.6590512    6.013842    4.520679    3.604186    3.738056    3.233470
## 4   1.1620616    6.923950    3.776700    4.599827    3.921958    5.326116
## 5   1.0666825    7.142722    4.982352    5.022859    5.914012    5.171385
## 6   1.2769005   10.531138    6.002749    7.937845    6.489814    6.081732
```

```
##   X2023.02.12 X2023.02.19 X2023.02.26 X2023.03.05 X2023.03.12 X2023.03.19
## 1    5.252223    5.632684    6.170898    5.577007    7.599948    6.412166
## 2    1.760620    3.479119    3.462271    2.485085    2.089156    2.156549
## 3    4.325023    4.170558    3.604186    4.252940    4.345619    3.923414
## 4    6.439758    7.287095    7.819706    7.577610    7.432352    7.359723
## 5    3.807651    5.900510    5.860003    5.009357    7.412768    6.170556
## 6    5.699978    7.793042    7.977337    6.410830    7.635075    8.688189
##   X2023.03.26 X2023.04.02 X2023.04.09 X2023.04.16 X2023.04.23 X2023.04.30
## 1    6.996777    6.653434    4.546977    5.215105    5.725479    5.753318
## 2    2.746230    2.897862    2.232365    1.389963    1.920676    1.819588
## 3    4.613358    3.799842    2.018344    3.212875    4.283833    4.716335
## 4    8.715462    8.328108    8.037593    6.996579    8.110222    8.400737
## 5    6.373090    7.048206    5.846500    6.211063    6.589127    6.008528
## 6    9.543844    8.806664    9.267401    8.477566    8.082648    7.319141
##   X2023.05.07 X2023.05.14 X2023.05.21 X2023.05.28 X2023.06.04 X2023.06.11
## 1    6.736950    7.405077    4.825364    4.639773    4.194355   3.2385613
## 2    2.661990    2.678838    2.325029    2.737806    1.710076   0.5307131
## 3    5.272410    4.386810    2.924540    3.367340    1.966856   2.1522141
## 4    8.884929    7.504981    7.311304    8.763881    7.747077   7.9407542
## 5    5.887007    6.332584    6.062537    5.198389    3.564610   2.4574205
## 6    8.819828    8.872484    7.740386    7.121682    5.805290   6.7662561
##   X2023.06.18 X2023.06.25 X2023.07.02 X2023.07.09 X2023.07.16 X2023.07.23
## 1    4.482020   4.6212136   5.1872659   5.1315886   4.6768909    2.459080
## 2    0.463321   0.9182179   0.4212009   0.2695686   0.8087057    0.943490
## 3    2.728884   2.1522141   1.9874513   2.7288838   2.3375722    2.265488
## 4    7.529191   8.1344312   9.5385889   8.8849293   8.5459946    8.473366
## 5    3.888665   3.7806469   2.7004620   2.1198627   3.5376053    3.821154
## 6    7.095354   7.9115174   5.8184537   4.3177666   5.8842733    5.884273
##   X2023.07.30 X2023.08.06 X2023.08.13 X2023.08.20 X2023.08.27 X2023.09.03
## 1   2.7189068   2.7467454    2.820982    4.194355    3.591184    3.841732
## 2   0.7750096   0.7076175    2.004916    2.013340    1.718500    1.297299
## 3   3.1098978   2.9863257    2.605312    2.574419    2.687693    2.203702
## 4   9.1754447   7.9407542    7.771287    7.093418    7.553400    8.521785
## 5   4.5097716   4.3612462    2.497927    3.834656    5.846500    6.926685
## 6   6.2002075   6.9505510    7.845698    6.687273    6.331847    6.766256
##   X2023.09.10 X2023.09.17 X2023.09.24 X2023.10.01 X2023.10.08 X2023.10.15
## 1    4.695450    4.584095    4.435623    5.632684    5.512050    5.706920
## 2    1.575291    1.389963    1.861708    2.333453    2.316605    2.569325
## 3    2.286084    2.450847    3.593889    2.873051    2.842158    3.130493
## 4    9.804895    9.151235    9.320702   10.143829   10.095410    8.739672
## 5    5.954519    4.563781    5.319910    6.616132    5.535947    5.468436
## 6    7.924681    6.239699    5.410372    5.489356    6.134388    6.015913
##   X2023.10.22 X2023.10.29 X2023.11.05 X2023.11.12 X2023.11.19 X2023.11.26
## 1    4.491300    5.066632    6.013145    6.690552    5.911070    4.546977
## 2    2.139701    2.451389    2.855742    2.923134    3.041070    2.813622
## 3    2.924540    3.315851    3.089302    3.913116    3.738056    3.460019
## 4    9.441750    8.013383    6.923950    8.231270    9.659637    8.836510
## 5    4.860832    5.333413    5.981523    5.887007    6.251570    5.063366
## 6    5.397208    5.581503    5.805290    6.621453    7.253321    5.015454
##   X2023.12.03 X2023.12.10 X2023.12.17 X2023.12.24 X2023.12.31 X2024.01.07
## 1    4.407784    6.634875    6.653434   2.5611545   1.0857068    4.083000
## 2    2.392421    3.133735    2.771502   0.9856101   0.2021764    2.089156
## 3    4.057284    3.511507    3.892521   2.6156094   0.9061954    1.637330
## 4    8.231270    9.514379   10.095410   5.7618887   2.7841059    7.069208
```

```
## 5    6.062537    4.928343    6.602630   3.2540568   1.9713373    6.859174
## 6    5.120766    8.043157    8.990959   4.9891266   2.0535719    7.529764
##   X2024.01.14 X2024.01.21 X2024.01.28 X2024.02.04 X2024.02.11 X2024.02.18
## 1    3.953086    4.435623    5.140868    3.943807    4.714009    3.823173
## 2    3.083191    3.108463    2.939982    2.737806    3.378031    2.569325
## 3    2.677395    3.099600    2.924540    3.305554    4.294130    4.129368
## 4    7.698658    8.667043   10.192249   10.821699   12.129018   10.095410
## 5    6.481109    7.493782    7.439773    4.725809    4.253228    3.888665
## 6    6.318683    5.436700    6.805748    6.147552    6.726764    7.714059
##   X2024.02.25 X2024.03.03 X2024.03.10 X2024.03.17 X2024.03.24 X2024.03.31
## 1    4.584095    5.038793    4.751127    4.871761    3.999484    4.482020
## 2    3.285367    2.400845    2.333453    3.066343    2.712534    2.249213
## 3    2.811265    2.965730    3.109898    3.573293    2.780372    3.264363
## 4    9.369122   10.192249    9.635427    8.909139   10.168039   10.264877
## 5    3.402582    4.671799    3.605117    4.685302    5.076869    3.645624
## 6    6.239699    4.844323    4.146636    4.225619    5.568339    4.844323
##   X2024.04.07 X2024.04.14 X2024.04.21 X2024.04.28 X2024.05.05 X2024.05.12
## 1    4.751127    5.066632    4.537698    3.405593    3.507668    3.025132
## 2    2.164973    2.855742    2.948406    2.889438    2.181821    2.544053
## 3    3.923414    3.387935    3.696865    3.315851    2.831861    2.471442
## 4    8.763881    8.497575    9.732266   10.071200    7.892335    7.626029
## 5    5.306408    5.400924    3.281061    3.780647    5.076869    2.389909
## 6    6.739928    8.517058    6.042240    3.988668    3.909685    5.318225
##   X2024.05.19 X2024.05.26 X2024.06.02 X2024.06.09 X2024.06.16 X2024.06.23
## 1    4.342827    3.711818    4.500580    4.268591    3.841732    3.331357
## 2    2.965254    3.386455    2.080732    2.341877    2.822046    2.763078
## 3    2.121321    2.172809    1.657926    2.234595    2.347870    2.430251
## 4   10.434345   10.579602   10.216458    7.553400    7.335514    6.827112
## 5    2.551937    4.172214    4.037191    5.279403    4.887836    4.050693
## 6    6.516142    3.646407    4.883815    4.989127    7.582419    6.279191
##   X2024.06.30 X2024.07.07 X2024.07.14 X2024.07.21 X2024.07.28 X2024.08.04
## 1    3.071530    2.310607    2.394123    3.080809    3.544786    3.201443
## 2    2.973678    2.392421    1.432083    1.667956    1.710076    1.769044
## 3    2.780372    2.162512    2.502335    1.760902    1.688819    1.894772
## 4    8.788091    9.102816   11.233262    9.417541    7.747077    7.698658
## 5    3.348573    3.510601    4.523274    3.861661    4.266730    2.160370
## 6    6.055404    4.475734    5.699978    6.094896    5.910601    5.713142
##   X2024.08.11 X2024.08.18 X2024.08.25 X2024.09.01 X2024.09.08 X2024.09.15
## 1    3.108648    3.043691    2.431241    3.303518    2.718907    1.967264
## 2    1.819588    1.255179    1.802740    2.097580    2.392421    2.308181
## 3    2.327275    1.935963    2.584716    2.667098    2.718586    3.490912
## 4    9.030187    3.921958    0.000000    2.493591    4.793504    6.028195
## 5    3.186545    3.065024    2.349402    2.551937    2.713964    4.766316
## 6    5.555175    5.054946    4.673192    4.857487    4.752176    5.397208
##   X2024.09.22 X2024.09.29 X2024.10.06 X2024.10.13 X2024.10.20 X2024.10.27
## 1    3.238561    3.257120    3.841732    4.203634    3.182884    3.127207
## 2    2.232365    2.805198    3.344335    2.518781    2.619870    2.417693
## 3    3.326149    3.058409    3.614484    3.357042    3.398233    2.358168
## 4    6.754483    7.650239    9.417541    8.788091    8.086012    6.633435
## 5    4.712306    4.266730    5.360417    3.834656    3.051522    2.754471
## 6    6.634617    6.990043    6.739928    6.042240    5.568339    5.173422
##   X2024.11.03 X2024.11.10 X2024.11.17 X2024.11.24 X2024.12.01 X2024.12.08
## 1    2.988014    3.359195    3.758216    3.479830    4.519139    4.120118
## 2    2.754654    3.049494    2.510357    3.184279    4.009833    3.765536
```

```
## 3    2.172809    3.367340    2.780372    3.027516    4.201451    4.180856
## 4    9.030187   10.942747    9.635427    9.829104    8.715462   10.313297
## 5    2.970508    4.820325    3.186545    3.389080    5.009357    4.307237
## 6    4.673192    6.239699    6.502978    6.108060    5.739470    5.502520
##   X2024.12.15 X2024.12.22 X2024.12.29 X2025.01.05 X2025.01.12 X2025.01.19
## 1    4.816084    4.639773    1.503286    3.925248    5.428534    4.426343
## 2    3.597056    2.965254    1.120394    1.794316    3.217975    3.201127
## 3    3.810140    2.193405    1.019470    3.490912    3.470316    3.655675
## 4    8.763881   10.192249    4.987181   10.579602   11.765874   11.160633
## 5    3.537605    4.833827    1.741798    5.292906    7.264243    7.021201
## 6    5.660487    5.278733    2.566965    7.174338    8.003665    8.227452
##   X2025.01.26 X2025.02.02 X2025.02.09 X2025.02.16 X2025.02.23 X2025.03.02
## 1    3.544786    4.361386    3.971645    4.444902    4.482020    4.500580
## 2    3.504391    4.085649    4.523698    3.807656    3.748688    2.577749
## 3    5.045861    3.851330    4.242642    2.512633    3.521805    2.625907
## 4    8.521785    9.514379    9.223864    8.449156   10.385925    9.974362
## 5    6.494611    6.332584    4.914841    4.212721    4.509772    4.725809
## 6    8.477566    9.635991    9.754467    7.213829    7.450780    8.477566
##   X2025.03.09 X2025.03.16 X2025.03.23 X2025.03.30 X2025.04.06 X2025.04.13
## 1    3.600464    3.117927    2.635391    3.368475    2.440520    3.340636
## 2    1.785892    1.912252    1.069850    2.603022    3.066343    1.482627
## 3    3.027516    2.306679    1.946261    1.348995    1.606437    1.657926
## 4    8.884929    7.892335    6.270291    9.320702    7.722868    7.601820
## 5    3.497098    2.092858    3.834656    4.739311    4.104702    2.308895
## 6    6.608289    6.213371    4.041324    5.844782    4.936471    4.436242
##   X2025.04.20 X2025.04.27 X2025.05.04 X2025.05.11 X2025.05.18 X2025.05.25
## 1    3.962366    3.860291    2.421961    2.394123    2.691068    3.071530
## 2    2.367149    1.272027    1.339419    1.196211    1.010882    1.516323
## 3    1.668223    1.935963    1.781498    2.008047    1.832986    2.399358
## 4    8.739672    9.611218    8.400737    8.497575    9.248074    8.521785
## 5    2.281890    2.646453    2.106360    2.673457    2.362904    2.200877
## 6    4.778504    5.515683    4.594209    5.423536    6.266027    4.660029
##   X2025.06.01 X2025.06.08 X2025.06.15 X2025.06.22 X2025.06.29 X2025.07.06
## 1    2.783864    1.679598    2.310607    3.683980    2.728186    2.635391
## 2    1.828012    1.945948    1.760620    1.566867    1.263603    1.592139
## 3    2.121321    2.275786    1.585842    1.802093    1.853581    1.740307
## 4    8.473366    8.134431    8.667043    6.585016    9.659637    8.933349
## 5    1.579770    2.106360    1.755300    2.470923    3.429587    2.970508
## 6    5.792126    6.121224    6.423994    5.910601    6.410830    5.607831
##   X2025.07.13 X2025.07.20 X2025.07.27 X2025.08.03 X2025.08.10 X2025.08.17
## 1    1.744555    1.985823    1.818791    2.022941    1.865189    2.524036
## 2    1.499475    1.524747    1.693228    0.951914    1.330995    1.853284
## 3    2.069833    2.471442    2.162512    1.905070    1.997749    2.337572
## 4    6.657645    7.795497    8.594414    9.441750    6.778693    5.568212
## 5    3.335071    2.470923    3.254057    3.159541    2.308895    2.943504
## 6    4.594209    7.069026    7.134846    6.937387    6.094896    7.740386
##   X2025.08.24 X2025.08.31 X2025.09.07 X2025.09.14 X2025.09.21 X2025.09.28
## 1    1.429050    2.570434    2.533316    1.800232    1.967264    1.707436
## 2    1.861708    1.625835    1.533171    1.743772    1.769044    1.886980
## 3    2.842158    2.564121    2.368465    3.243768    3.573293    2.419954
## 4    9.369122    9.199654    9.272283    7.964964    8.884929    9.320702
## 5    2.700462    3.591615    4.644795    3.132536    3.146038    4.253228
## 6    8.240615    7.951009    7.134846    8.582877    7.253321    4.080816
##   X2025.10.05 X2025.10.12 X2025.10.19 X2025.10.26 X2025.11.02 X2025.11.09
```

```
## 1    2.421961    2.468359    1.735275    1.716716    2.273489    2.746745
## 2    2.375573    1.642683    1.246755    1.280451    1.457355    2.063884
## 3    2.131619    2.770075    2.955433    2.862754    2.728884    4.458893
## 4   10.700651    6.972370    4.502989    3.728281    4.260893    4.236683
## 5    4.536776    5.211892    5.670970    4.320739    5.076869    6.197560
## 6    5.291897    6.884731    4.778504    5.568339    4.804832    6.476650
##   X2025.11.16
## 1    2.895218
## 2    1.996492
## 3    3.665972
## 4    4.067216
## 5    2.808481
## 6    5.620995
```

Reformat Dataframe to Matrix Formatting as matrix with ~ rows (regions) x columns (weeks)

```
uecMat = as.matrix(uecDf)
```

Model 3 Specification

Model 3 adds region-specific New Year effects plus the Mid West reset to the baseline AR(1) seasonal structure.

```
model=
"
model{
  # Iterate through the regions
  for(i in 1:I){
    #------------
    #Likelihood
    #------------
    # Set first data point in region i to normal
    y[i,1] ~ dnorm(mu[i,1], tau[i])
    # Set second on data points to the AR1 model with a mean with a seasonal component
    for(t in 2:T){
      y[i,t] ~ dnorm(mu[i,t] +
      (phi * (y[i,t-1] - mu[i,t-1])), tau[i])
    }
    # Assign mean to each time point
    # Beta  ~ cosine coeffecient scaler
    # Gamma ~ sine coeffecient scaler
    for(t in 1:T){
      mu[i,t] <- alpha[i] +
                 beta[i]      * cos((2 * pi) * (t/52)) +
                 gamma[i]     * sin((2 * pi) * (t/52)) +

                 delta_pre[i]   * ny_pre[t]  +
                 delta_mid[i]   * ny_mid[t]  +
                 delta_post[i]  * ny_post[t] +

                 sigma_pre  * fr_pre[t] * mw[i] +
                 sigma_mid  * fr_mid[t] * mw[i] +
                 sigma_post * fr_post[t]* mw[i]
    }
    #Full conditional  mean used in the likelihood
```

```
    fullmod[i,1] <- mu[i,1]
    for(t in 2:T){
      fullmod[i,t] <- mu[i,t] + phi * (y[i,t-1] - mu[i,t-1])
    }
    #Residuals
    for(t in 1:T){
      resid[i,t] <- y[i,t] - fullmod[i,t]
    }
    #Uninformative Priors
    alpha[i] ~ dnorm(0, 0.001)
    beta[i]  ~ dnorm(0, 0.001)
    gamma[i] ~ dnorm(0, 0.001)
    tau[i]   ~ dgamma(0.001, 0.001)
    delta_pre[i]  ~ dnorm(0, 0.001)
    delta_mid[i]  ~ dnorm(0, 0.001)
    delta_post[i] ~ dnorm(0, 0.001)
  }
  phi ~ dunif(-1, 1)
  sigma_pre  ~ dnorm(0, 0.001)
  sigma_mid  ~ dnorm(0, 0.001)
  sigma_post ~ dnorm(0, 0.001)
}
"
# New year indicators: pre, mid, post
week_mod  = (1:n.weeks) %% 52
# as numeric makes a mask
ny_pre    = as.numeric(week_mod == 0)  # week before dip
ny_mid    = as.numeric(week_mod == 1)  # the dip
ny_post   = as.numeric(week_mod == 2)  # week after dip
fr_pre  = as.numeric((1:n.weeks) == 86)  # week before reset
fr_mid  = as.numeric((1:n.weeks) == 87)  # the reset
fr_post = as.numeric((1:n.weeks) == 88)  # week after reset
mw      = as.numeric(regions == "HSE Mid West") # 1 only for Mid West
```

Fit

```
# fit model
exp_jags = jags.model(textConnection(model),
                      list(y  = uecMat,
                           I  = n.region,
                           T  = n.weeks,
                           pi = pi,
                           ny_pre    = ny_pre,
                           ny_mid    = ny_mid,
                           ny_post   = ny_post,
                           fr_pre    = fr_pre,
                           fr_mid    = fr_mid,
                           fr_post   = fr_post,
                           mw        = mw), n.chains = 4)
```

```
## Compiling model graph
##    Resolving undeclared variables
##    Allocating nodes
```

```
## Graph information:
##    Observed stochastic nodes: 906
##    Unobserved stochastic nodes: 46
##    Total graph size: 6928
##
## Initializing model
```

```
update(exp_jags, n.iter = 10000)
# Parameters to watch
model_parameters =  c("alpha","beta","gamma",
                      "tau","phi",
                      "delta_pre","delta_mid","delta_post",
                      "sigma_pre","sigma_mid","sigma_post")
# Sample
exp_sim=coda.samples(model=exp_jags,
                     variable.names=model_parameters,
                     n.iter=20000)

gelman_results = gelman.diag(exp_sim, multivariate = FALSE)
gelman_results
```

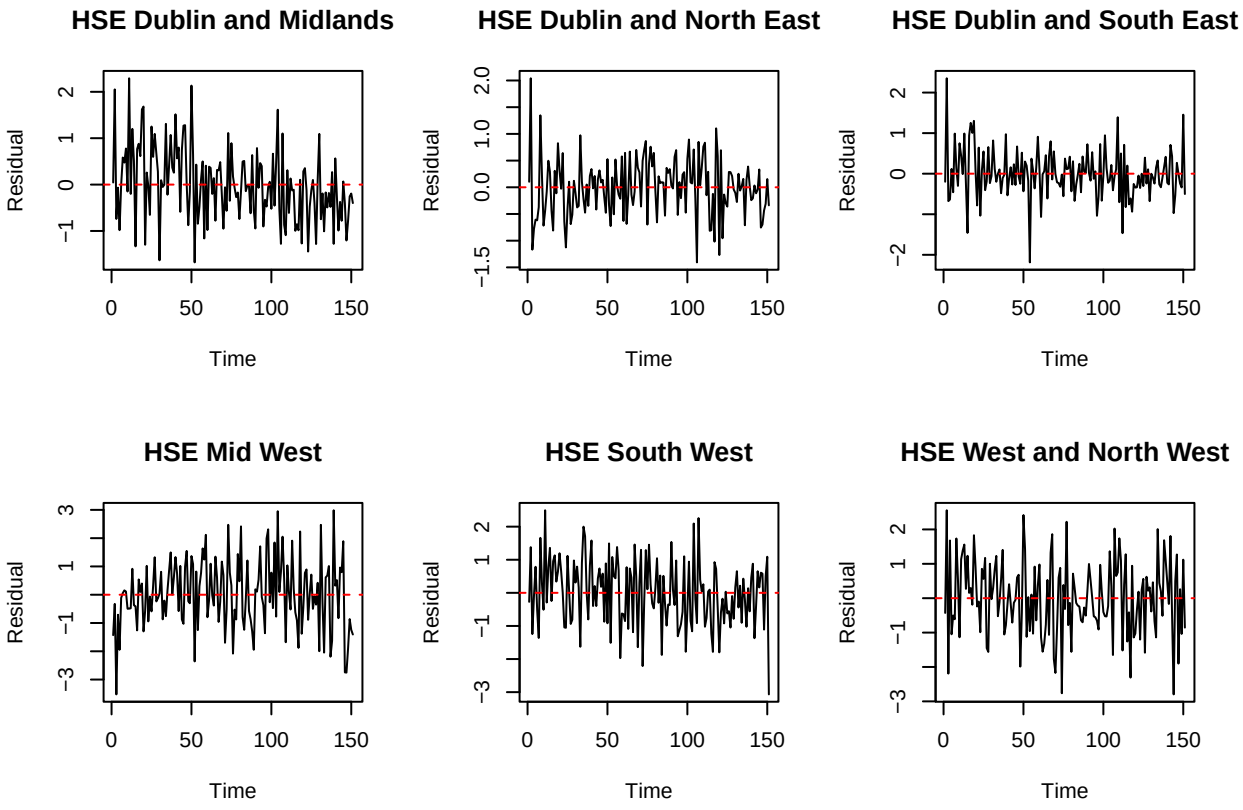
```
## Potential scale reduction factors:
##
##               Point est. Upper C.I.
## alpha[1]               1          1
## alpha[2]               1          1
## alpha[3]               1          1
## alpha[4]               1          1
## alpha[5]               1          1
## alpha[6]               1          1
## beta[1]                1          1
## beta[2]                1          1
## beta[3]                1          1
## beta[4]                1          1
## beta[5]                1          1
## beta[6]                1          1
## delta_mid[1]           1          1
## delta_mid[2]           1          1
## delta_mid[3]           1          1
## delta_mid[4]           1          1
## delta_mid[5]           1          1
## delta_mid[6]           1          1
## delta_post[1]          1          1
## delta_post[2]          1          1
## delta_post[3]          1          1
## delta_post[4]          1          1
## delta_post[5]          1          1
## delta_post[6]          1          1
## delta_pre[1]           1          1
## delta_pre[2]           1          1
## delta_pre[3]           1          1
## delta_pre[4]           1          1
## delta_pre[5]           1          1
## delta_pre[6]           1          1
## gamma[1]               1          1
```

```
## gamma[2]                1          1
## gamma[3]                1          1
## gamma[4]                1          1
## gamma[5]                1          1
## gamma[6]                1          1
## phi                     1          1
## sigma_mid               1          1
## sigma_post              1          1
## sigma_pre               1          1
## tau[1]                  1          1
## tau[2]                  1          1
## tau[3]                  1          1
## tau[4]                  1          1
## tau[5]                  1          1
## tau[6]                  1          1
```

Fitted Values And Residuals

```
# Posterior summary for monitored parameters
jags_sum <- summary(exp_sim)
df.stats = cbind(
  as.data.frame(jags_sum$statistics),
  as.data.frame(jags_sum$quantiles)
) %>% rownames_to_column(var = "index")

# Posterior means
alpha_mean <- df.stats %>%
  filter(grepl("^alpha\\[", index)) %>%
  arrange(as.integer(str_extract(index, "\\d+"))) %>%
  pull(Mean)
beta_mean <- df.stats %>%
  filter(grepl("^beta\\[", index)) %>%
  arrange(as.integer(str_extract(index, "\\d+"))) %>%
  pull(Mean)
gamma_mean <- df.stats %>%
  filter(grepl("^gamma\\[", index)) %>%
  arrange(as.integer(str_extract(index, "\\d+"))) %>%
  pull(Mean)
delta_pre_mean <- df.stats %>%
  filter(grepl("^delta_pre\\[", index)) %>%
  arrange(as.integer(str_extract(index, "\\d+"))) %>%
  pull(Mean)
delta_mid_mean <- df.stats %>%
  filter(grepl("^delta_mid\\[", index)) %>%
  arrange(as.integer(str_extract(index, "\\d+"))) %>%
  pull(Mean)
delta_post_mean <- df.stats %>%
  filter(grepl("^delta_post\\[", index)) %>%
  arrange(as.integer(str_extract(index, "\\d+"))) %>%
  pull(Mean)
phi_mean <- df.stats %>% filter(index == "phi") %>% pull(Mean)
sigma_pre_mean <- df.stats %>% filter(index == "sigma_pre") %>% pull(Mean)
sigma_mid_mean <- df.stats %>% filter(index == "sigma_mid") %>% pull(Mean)
sigma_post_mean <- df.stats %>% filter(index == "sigma_post") %>% pull(Mean)
```

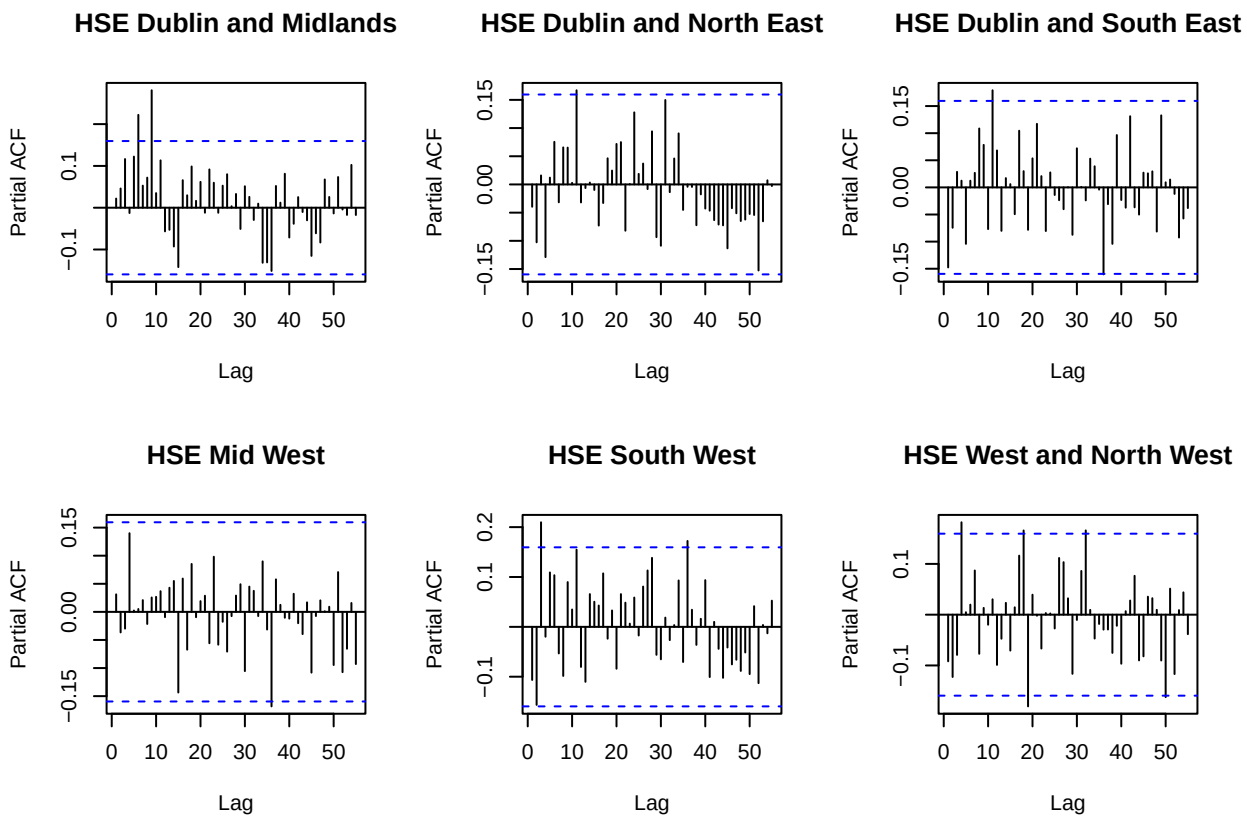
```r
# Reconstruct mu and AR(1) fitted values from posterior means
matrix_mu <- matrix(0, nrow = n.region, ncol = n.weeks)
matrix_fullmod <- matrix(0, nrow = n.region, ncol = n.weeks)

for (i in 1:n.region) {
  for (t in 1:n.weeks) {
    matrix_mu[i, t] <- alpha_mean[i] +
      beta_mean[i] * cos((2 * pi) * (t / 52)) +
      gamma_mean[i] * sin((2 * pi) * (t / 52)) +
      delta_pre_mean[i] * ny_pre[t] +
      delta_mid_mean[i] * ny_mid[t] +
      delta_post_mean[i] * ny_post[t] +
      sigma_pre_mean * fr_pre[t] * mw[i] +
      sigma_mid_mean * fr_mid[t] * mw[i] +
      sigma_post_mean * fr_post[t] * mw[i]
  }

  matrix_fullmod[i, 1] <- matrix_mu[i, 1]
  for (t in 2:n.weeks) {
    matrix_fullmod[i, t] <- matrix_mu[i, t] + phi_mean * (uecMat[i, t - 1] - matrix_mu[i, t - 1])
  }
}

# Save fitted values in long form
df_fitted = as.data.frame(t(matrix_fullmod))
colnames(df_fitted) = regions
df_fitted$time = seq_len(n.weeks)
df_fitted = df_fitted %>% pivot_longer(-time, names_to = "Region", values_to = "fitted")
if (save_output) write.csv(df_fitted, paste0(output_folder, "fitted.csv"), row.names = FALSE)

# Residual diagnostics
matrix_resid = t(uecMat - matrix_fullmod)
df_resid = as.data.frame(matrix_resid)
colnames(df_resid) = regions
df_resid$time = seq_len(nrow(df_resid))

# 2x3 residual plot (one per region)
par(mfrow = c(2, 3))
for (j in 1:6) {
  plot(df_resid$time, matrix_resid[, j], type = "l",
       main = regions[j], xlab = "Time", ylab = "Residual")
  abline(h = 0, col = "red", lty = 2)
}
```



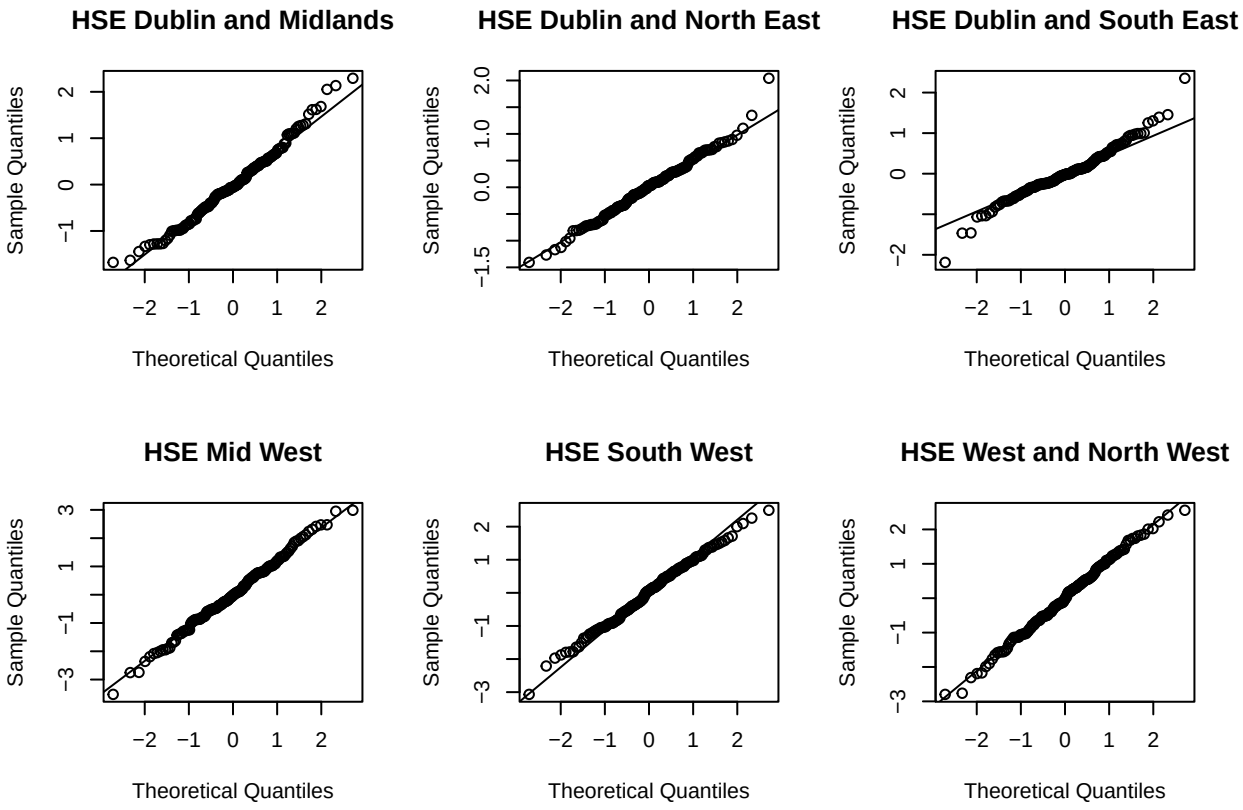

```
par(mfrow = c(1, 1))

# Residual outliers
df_outliers = df_resid %>%
  pivot_longer(-time, names_to="Region", values_to="Residual") %>%
  filter(abs(Residual) > 3)
df_outliers
```

```
## # A tibble: 2 x 3
##    time Region         Residual
##   <int> <chr>             <dbl>
## 1     3 HSE Mid West      -3.52
## 2   151 HSE South West    -3.07
```

```
# PACF by region
par(mfrow = c(2, 3))
for (j in 1:6) {
  pacf(matrix_resid[, j], lag.max = 55, main = regions[j])
}
```




```r
par(mfrow = c(1, 1))

# QQ by region
par(mfrow = c(2, 3))
for (j in 1:6) {
  qqnorm(matrix_resid[, j], main = regions[j]); qqline(matrix_resid[, j])
}
```




```
par(mfrow = c(1, 1))
```

DIC

```
dic_v3 = dic.samples(exp_jags, n.iter = 20000)
dic_v3
```

```
## Mean deviance:  2278
## penalty 46.76
## Penalized deviance: 2325
```

```
df_dic = data.frame(
  deviance = sum(dic_v3$deviance),
  penalty  = sum(dic_v3$penalty),
  DIC      = sum(dic_v3$deviance) + sum(dic_v3$penalty)
)
if (save_output) write.csv(df_dic, paste0(output_folder, "dic.csv"), row.names = FALSE)
```

Save

```
if (save_output) {
  samples_with_chain = do.call(rbind, lapply(seq_along(exp_sim), function(ch) {
    df = as.data.frame(exp_sim[[ch]])
    df$chain = ch
    df
  }))
  write.csv(samples_with_chain,
            paste0(output_folder, "raw_samples.csv"),
```

```
            row.names = FALSE)
}
```

Budget-Scaled Variant

```
uecDf    = read.csv('../data/wide_weekly_scaledPerBudgetPop.csv')
regions  = uecDf$Region # saving region names
n.region = length(uecDf$Region) # number of regions
uecDf    = uecDf %>% select(-Region) # remove region col
n.weeks  = length(names(uecDf))
uecMat   = as.matrix(uecDf)

# New year indicators: pre, mid, post
week_mod  = (1:n.weeks) %% 52
# as numeric makes a mask
ny_pre    = as.numeric(week_mod == 0)  # week before dip
ny_mid    = as.numeric(week_mod == 1)  # the dip
ny_post   = as.numeric(week_mod == 2)  # week after dip
fr_pre  = as.numeric((1:n.weeks) == 86)  # week before reset
fr_mid  = as.numeric((1:n.weeks) == 87)  # the reset
fr_post = as.numeric((1:n.weeks) == 88)  # week after reset
mw      = as.numeric(regions == "HSE Mid West") # 1 only for Mid West

# fit model
exp_jags = jags.model(textConnection(model),
                      list(y  = uecMat,
                           I  = n.region,
                           T  = n.weeks,
                           pi = pi,
                           ny_pre    = ny_pre,
                           ny_mid    = ny_mid,
                           ny_post   = ny_post,
                           fr_pre    = fr_pre,
                           fr_mid    = fr_mid,
                           fr_post   = fr_post,
                           mw        = mw), n.chains = 4)
```

```
## Compiling model graph
##    Resolving undeclared variables
##    Allocating nodes
## Graph information:
##    Observed stochastic nodes: 906
##    Unobserved stochastic nodes: 46
##    Total graph size: 6928
##
## Initializing model
```

```
update(exp_jags, n.iter = 10000)

# Parameters to watch
model_parameters =  c("alpha","beta","gamma",
                      "tau","phi",
                      "delta_pre","delta_mid","delta_post",
                      "sigma_pre","sigma_mid","sigma_post")
```

```r
# Sample
exp_sim=coda.samples(model=exp_jags,
                     variable.names=model_parameters,
                     n.iter=20000)

# postfix
write_postfix = "_scaledPerBudgetPop"

# Check convergence
gelman_results = gelman.diag(exp_sim,
  multivariate = FALSE)# added when there are  alot of variables so it convergees
write.csv(as.matrix(gelman_results),
          paste0(output_folder, "gelman", write_postfix,".csv"),
          row.names = FALSE)

# Check Deviance
dic_v3 = dic.samples(exp_jags, n.iter = 20000)
df_dic = data.frame(
  deviance = sum(dic_v3$deviance),
  penalty  = sum(dic_v3$penalty),
  DIC      = sum(dic_v3$deviance) + sum(dic_v3$penalty)
)
write.csv(df_dic, paste0(output_folder, "dic", write_postfix,".csv"), row.names = FALSE)
# Write raw samples
samples_with_chain = do.call(rbind, lapply(seq_along(exp_sim), function(ch) {
  df = as.data.frame(exp_sim[[ch]])
  df$chain = ch
  df
}))
write.csv(samples_with_chain,
          paste0(output_folder, "raw_samples", write_postfix,".csv"),
          row.names = FALSE)
```